



PUSL3190 Computing Project

Abstract

Stock Price Indicator And Prediction Project

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Research Abstract

Predicting stock prices is challenging because the market is constantly changing and influenced by many complex factors. While advanced AI models like Long Short-Term Memory (LSTM) are excellent at identifying these patterns, they are usually hidden away in complex coding tools that are hard for the average person to use. This often leaves individual investors at a disadvantage, relying on outdated methods or expensive professional software. This project aims to bridge that gap by building a user-friendly web platform that brings high-tech financial forecasting to everyone. The system is built using a "split" architecture to ensure it stays fast and responsive. On the backend, a specialized Python service fetches data from Yahoo Finance and uses an LSTM model to predict future price trends based on historical data and technical indicators like RSI and MACD. On the frontend, a modern Next.js interface presents these predictions through clear, interactive charts. By separating the heavy data processing from the web interface, the application provides a smooth experience that can easily grow as more users join. The final result is a practical, all-in-one tool that helps everyday investors and researchers make more informed, data-driven decisions without needing to be experts in programming or data science.

Keywords: Stock Price Prediction, Long Short-Term Memory (LSTM), Financial Forecasting, Next.js, Microservice Architecture, Technical Indicators (RSI, MACD), Data Visualization Python (FastAPI).